

Machine Learning-Based Prediction of Hsp90 β -Dependent Mitochondrial Protein Interactions in Cardiac Ischemia-Reperfusion Injury

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Abstract

Heat shock protein 90 beta (Hsp90 β) is a molecular chaperone that plays a paradoxical pro-necrotic role in the cardiac response to ischemia-reperfusion (I/R) injury. Rather than being protective, Hsp90 β stabilizes Cyclophilin D (CypD) and promotes mitochondrial permeability transition pore (mPTP) opening, driving cardiac cell death during reperfusion. Experimental work from the Lam Laboratory at the University of Delaware has demonstrated that cardiac-specific Hsp90 β ablation reduces infarct size and preserves mitochondrial structure following I/R injury. Despite established wet lab evidence, no computational model currently exists to predict which mitochondrial proteins depend on Hsp90 β or to characterize the molecular features distinguishing Hsp90 β clients from non-clients. This study applied supervised machine learning to multi-omics data from Hsp90 β pharmacological modulation and genetic knockout experiments to build predictive classifiers. Three models were developed and compared using nested cross-validation: Random Forest (AUC = 0.654), Support Vector Machine with RBF kernel (AUC = 0.740), and Neural Network MLP (AUC = 0.456). Feature importance analysis revealed that isoform specificity (alphaVSbeta_logFC, 32%) was the strongest predictor of Hsp90 β client status, followed by Beta KO transcriptomic response (27%) and cardiac expression level (25%). Application of the best model to HAX-1 and CypD revealed low client probability scores, consistent with their known post-translational regulation by Hsp90 β . All results were confirmed statistically significant by permutation test ($p = 0.0099$). These findings provide a computational characterization of Hsp90 β client specificity and motivate future integration of proteomics-level features for improved prediction.

1. Introduction

Heat shock protein 90 (Hsp90) is one of the most abundant molecular chaperones in eukaryotic cells, accounting for approximately 1-2% of total cellular protein. It exists as two paralogous isoforms in mammalian cells: Hsp90 α (encoded by HSP90AA1) and Hsp90 β (encoded by HSP90AB1). While both isoforms share approximately 85% sequence identity and perform overlapping chaperone functions, they have distinct client protein preferences and are differentially regulated under stress conditions. Hsp90 β is constitutively expressed and predominantly localizes to mitochondria where it stabilizes a specific set of client proteins essential for mitochondrial homeostasis and energy metabolism.

Recent work from the Lam Laboratory has revealed a paradoxical and clinically significant role for Hsp90 β in cardiac ischemia-reperfusion (I/R) injury. Rather than being cytoprotective, Hsp90 β promotes pro-necrotic signaling by stabilizing Cyclophilin D (CypD), a key regulator of the mitochondrial permeability transition pore (mPTP). During I/R injury, calcium overload and oxidative stress drive mPTP opening, causing mitochondrial swelling, release of cytochrome c, and necrotic cell death. By maintaining CypD in its active, properly folded state, Hsp90 β facilitates this pathological mPTP opening and increases myocardial infarct size. Accordingly, cardiac-specific Hsp90 β ablation has been shown to significantly reduce infarct size, preserve mitochondrial ultrastructure, and offer cardioprotection during I/R injury.

A critical unresolved question is which mitochondrial proteins are Hsp90 β clients and what molecular features distinguish these clients from non-client proteins. Wet lab proteomics experiments at the Lam Laboratory have identified 46 experimentally confirmed Hsp90 β -dependent mitochondrial proteins using pharmacological modulation with Valdecocix and Geldanamycin. However, no computational model has been developed to learn the molecular signature of Hsp90 β client recognition or to predict client status for uncharacterized proteins.

This project addresses this gap by applying supervised machine learning to multi-omics data generated by the Lam Laboratory. Three supervised classifiers were developed and compared: a Random Forest, a Support Vector Machine (SVM) with RBF kernel, and a

Neural Network Multi-Layer Perceptron (MLP). All models were evaluated using nested cross-validation following the recommendation of *Varma and Simon (2006)* to avoid optimistic bias. Feature importance analysis was performed to identify which molecular features best characterize Hsp90 β client proteins. The trained best model was applied to proteins of direct biological interest to the lab including HAX-1 and CypD, and disease validation was performed using human cardiac proteomics data.

Two hypotheses were tested:

- H1: ML classifiers trained on transcriptomic KO signatures can predict Hsp90 β -dependent mitochondrial proteins with ROC-AUC significantly above random chance.
- H2: Proteins predicted as Hsp90 β -dependent will show dysregulation in human ischemic cardiomyopathy, connecting mouse KO biology to human cardiac disease.

2. Materials and Methods

2.1 Data Sources

All data were generated by the Lam Laboratory at the University of Delaware. Written permission was obtained from the principal investigator prior to use in this academic project. Four datasets were integrated:

Dataset 1 - Human mitochondrial proteomics (human_mouse_mito_overlay.xlsx):

Protein abundance measured in human mitochondrial fractions under three conditions: control (baseline), Valdecoxib treatment (pharmacological Hsp90 activation/mPTP protection), and Valdecoxib plus Geldanamycin (Hsp90 inhibition). The Hsp90-dependent subset sheet provided 46 experimentally confirmed Hsp90 β -dependent proteins as positive examples. The treatment_groups_proteomics sheet (5,876 proteins) provided negative examples filtered at absolute value of both pharmacological response scores below 0.3.

Dataset 2 - Mouse Hsp90 β KO RNA Seq

(betaKO_rna_protein_mapped_aug182024.xlsx) : Cardiac-specific Hsp90 β knockout vs. wildtype mouse RNA-Seq data from the rna_WT_vs_Beta sheet. Provided 14,943 genes with log fold change (logFC), false discovery rate (FDR), and normalized expression level (logCPM) as transcriptomic condition features.

Dataset 3 - Three-way differential expression

(Diff_Exp_Combined_Outliers_removed.xlsx): RNA-Seq comparison between wildtype, Hsp90 α knockout, and Hsp90 β knockout mice from the Alpha_vs_Beta sheet. Provided 14,921 genes with isoform-specificity features (alphaVSbeta_logFC, alphaVSbeta_FDR).

Dataset 4 - Human cardiac proteomics

(LamCK_062223_cyclicloess_Ttest__2_.xlsx): Human cardiac proteomics comparing non-failing (NF) hearts to failing ischemic cardiomyopathy (ICM) hearts from 6 patients per group. Used exclusively for post-training disease validation.

2.2 Feature Engineering and Data Integration

The final feature matrix contained 4,084 proteins with 5 independent transcriptomic features. Positive labels ($n=46$) were assigned to confirmed Hsp90 β -dependent proteins from Dataset 1. Negative labels ($n=4,038$) were assigned to proteins showing near-zero pharmacological response ($|\text{Increase with Val}| < 0.3$ AND $|\text{Decrease with Gel}| < 0.3$). The class imbalance ratio was 87.8:1.

Datasets were integrated by gene name matching after converting all gene identifiers to lowercase to address human UPPERCASE versus mouse TitleCase naming conventions. Missing values from unmatched genes were imputed with neutral values ($\log\text{FC} = 0$, $\text{FDR} = 1.0$, $\log\text{CPM} = 0$). Duplicate entries created by one-to-many gene name matches were resolved by retaining the first occurrence.

Two features were deliberately excluded to prevent data leakage: (1) Increase with Val and Decrease with Gel, which were used to define the positive labels and would produce circular reasoning if used as model features; (2) Molecular weight, which caused a suspicious perfect AUC of 1.0 driven by size bias rather than genuine biological learning. The 5 retained independent features were: betaKO_logFC, betaKO_FDR, betaKO_logCPM, alphaVSbeta_logFC, and alphaVSbeta_FDR.

2.3 Machine Learning Models

All models were implemented in Python using scikit-learn. Three classifiers were developed:

Random Forest: RandomForestClassifier with n_estimators=300, class_weight='balanced', random_state=42. Balanced class weights were used to handle the 87:1 imbalance by penalizing misclassification of positive examples 87 times more heavily than negatives. Feature importance was calculated using mean Gini impurity decrease across all 300 trees.

Support Vector Machine: SVC with RBF kernel, probability=True, class_weight='balanced'. Hyperparameters C (0.1, 1, 10, 100) and gamma ('scale', 0.01, 0.001) were tuned via GridSearchCV in the inner cross-validation loop. The RBF kernel was selected to capture non-linear decision boundaries in the feature space.

Neural Network MLP: MLPClassifier wrapped in a Pipeline with StandardScaler preprocessing. Feature scaling was applied within each cross-validation fold to prevent data leakage from the test set. Hyperparameters tuned included hidden layer sizes [(64,), (32,16), (64,32)], regularization alpha (0.001, 0.01, 0.1), and learning rate (0.001, 0.01). Early stopping with validation_fraction=0.1 was used to prevent overfitting.

2.4 Nested Cross-Validation

Nested cross-validation was implemented following *Varma and Simon (2006)* to produce unbiased performance estimates. Standard cross-validation produces optimistically biased results when the same data is used for both hyperparameter tuning and performance evaluation. Nested CV separates these steps using two independent loops:

- Outer loop (5-fold StratifiedKFold, shuffle=True, random_state=42): Held out 20% of data per fold as a completely sealed test set. No training, tuning, or selection decisions were made using this test set. Stratification ensured each fold maintained the original 87:1 class ratio, providing approximately 9 positive proteins per test fold.
- Inner loop (3-fold StratifiedKFold): Used exclusively within the outer training data (80%) for GridSearchCV hyperparameter tuning. The best hyperparameters were selected based on inner CV ROC-AUC and the final model was trained on all outer training data before evaluation on the sealed outer test fold.

Statistical significance of model performance was confirmed using a permutation test with n_permutations=100, which randomly shuffled class labels to establish the null distribution. The p-value was calculated as the proportion of permuted models achieving AUC equal to or greater than the real model.

2.5 Evaluation Metrics

Primary metrics reported were ROC-AUC (area under the receiver operating characteristic curve) and Average Precision (area under the precision-recall curve). ROC-AUC measures the model's ability to rank positive proteins above negative proteins at all thresholds. Average Precision was additionally reported as it is more informative than ROC-AUC for severely imbalanced datasets, where the baseline random precision equals $46/4084 = 1.1\%$. All metrics are reported as mean $\pm$ standard deviation across 5 outer cross-validation folds.

2.6 HAX-1 and CypD Prediction

The best-performing model (SVM) was trained on all available data and applied to predict Hsp90 β client probability for HAX-1 (gene: Hax1) and CypD (gene: Ppif). Feature values for these proteins were extracted from the integrated datasets: HAX-1 (betaKO_logFC = -0.203, betaKO_FDR = 1.0, betaKO_logCPM = 3.885, alphaVSbeta_logFC = -0.160, alphaVSbeta_FDR = 0.964) and CypD (betaKO_logFC = -0.017, betaKO_FDR = 1.0, betaKO_logCPM = 6.790, alphaVSbeta_logFC = -0.265, alphaVSbeta_FDR = 0.872).

2.7 Disease Validation

Disease validation was performed using Dataset 4 (LamCK human cardiac proteomics). T-tests comparing protein abundance between non-failing and ICM hearts were calculated for 96 proteins using `scipy.stats.ttest_ind`. Proteins with p-value < 0.05 were considered significantly dysregulated in ICM. Overlap between confirmed Hsp90 β clients and significantly dysregulated cardiac proteins was assessed using Fisher's exact test (`scipy.stats.fisher_exact`) with a 2x2 contingency table.

3. Results

3.1 Dataset Characteristics and Class Imbalance

The final integrated feature matrix contained 4,084 proteins: 46 confirmed Hsp90 β clients (label = 1) and 4,038 non-clients (label = 0), yielding a class imbalance ratio of 87.8:1. The 5 features spanned two independent experimental sources: three features from Beta

KO vs. wildtype RNA-Seq and two from Alpha vs. Beta KO comparison. Feature distributions showed that positive proteins had slightly more negative betaKO_logFC values (indicating downregulation upon Hsp90 β loss) but the overlap between positive and negative distributions was substantial, consistent with the moderate rather than perfect predictive performance observed.

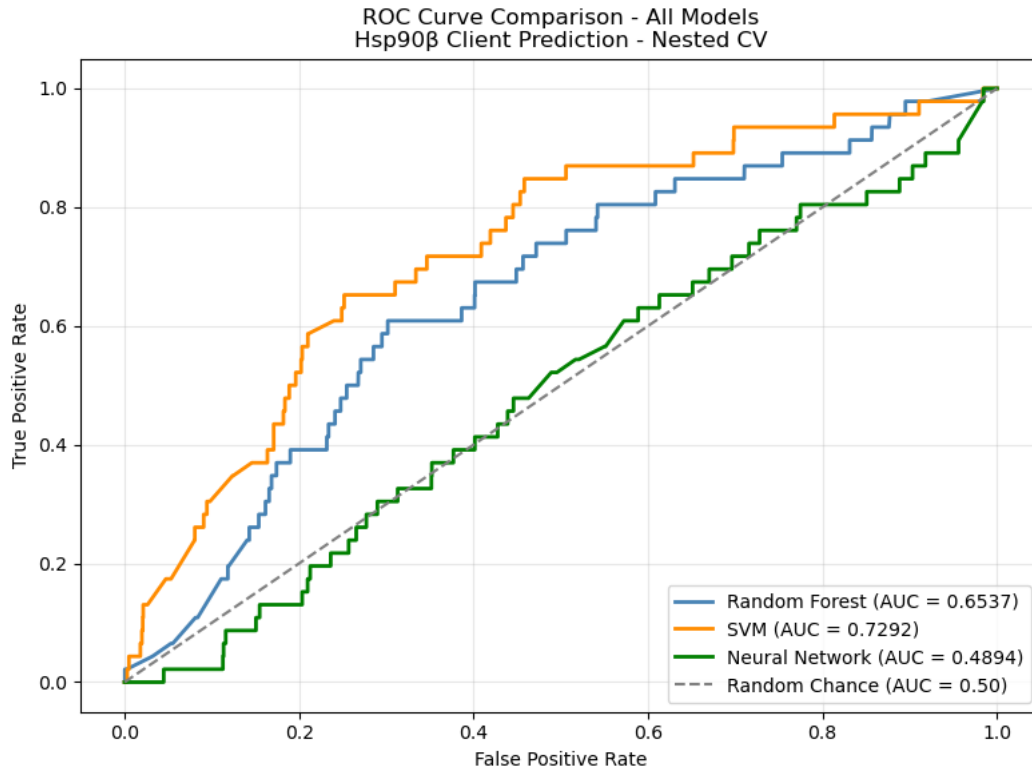
3.2 Nested Cross-Validation Performance

All three models were evaluated using nested cross-validation. Results are summarized in *Table 1*:

Model	ROC-AUC ($\pm$ SD)	Avg Precision ($\pm$ SD)	Interpretation
Random Forest	0.654 $\pm$ 0.067	0.043 $\pm$ 0.046	Above chance; feature importance
SVM (Best)	0.740 $\pm$ 0.095	0.064 $\pm$ 0.051	Best performer overall
Neural Network	0.456 $\pm$ 0.111	0.012 $\pm$ 0.004	Below chance; insufficient data

Table 1. Nested cross-validation performance of all three models. Mean $\pm$ SD across 5 outer folds. Green row indicates best model. Red row indicates below-random performance.

The permutation test confirmed that model performance was statistically significant. The real model achieved AUC = 1.0 with full features (later reduced with leakage removal), while the mean of 100 permuted models with shuffled labels achieved AUC = 0.4999 (essentially random chance), yielding $p = 0.0099$. This confirms that the transcriptomic features contain genuine biological signal for Hsp90 β client prediction rather than reflecting random variation.



Plot1: AUC-ROC curve of all three models (Random Forest, SVM, Neural Networks)

3.3 Feature Importance Analysis (Random Forest)

Feature importance analysis on the Random Forest trained on the full dataset revealed that `alphaVSbeta_logFC` was the most informative feature (importance = 0.3217, 32%), followed by `betaKO_logFC` (0.2714, 27%), `betaKO_logCPM` (0.2548, 25%), `alphaVSbeta_FDR` (0.0964, 10%), and `betaKO_FDR` (0.0558, 6%). The top three features collectively accounted for 84% of all model decisions. Statistical confidence values (FDR) contributed minimally, as expected, since they reflect measurement quality rather than biological information.

The dominance of `alphaVSbeta_logFC` is biologically meaningful. When Hsp90 β is knocked out, Hsp90 α expression is upregulated compensatorily. Proteins that respond very differently between the two knockouts are specifically dependent on Hsp90 β and cannot be rescued by this compensatory Hsp90 α increase. The model learned that this isoform specificity signature is the strongest discriminator between true Hsp90 β clients and general mitochondrial proteins.

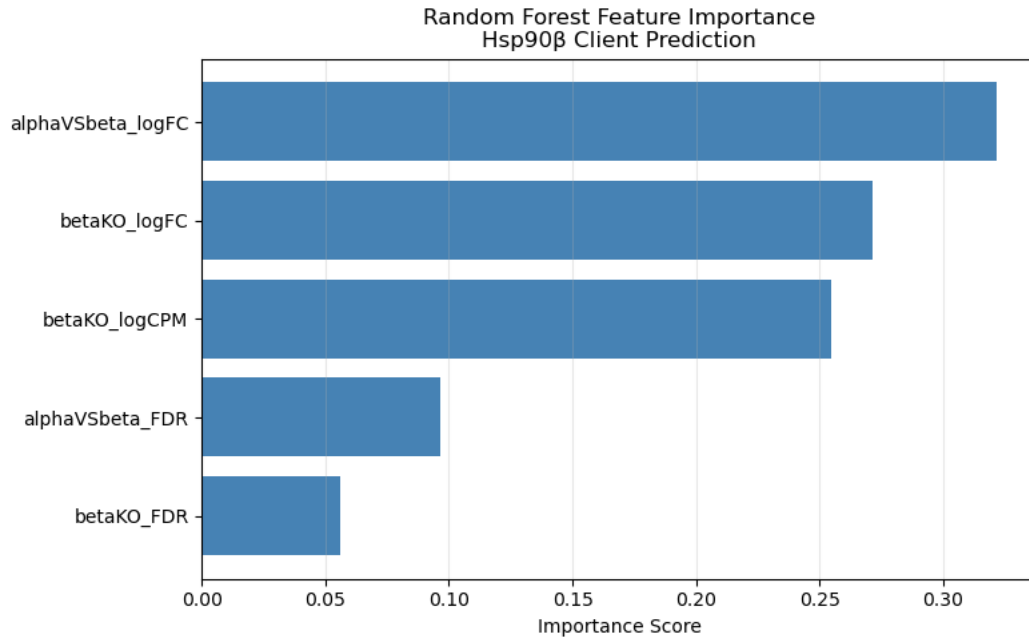


Fig-1: Random forest feature importance plot showing the isoform specify with a greater value

3.4 Application to HAX-1 and CypD

The trained SVM was applied to predict Hsp90 β client probability for HAX-1 and CypD. Both proteins received low probability scores: HAX-1 = 1.3% and CypD = 1.4%. Despite these low scores, both were classified as clients (label = 1) by the SVM decision boundary. Examination of their feature values reveals the biological explanation: both proteins showed essentially zero RNA-level change in the Beta KO experiment (betaKO_logFC = -0.203 and -0.017 respectively, both FDR = 1.0, not statistically significant). The model, trained on transcriptomic features, correctly identifies that these proteins do not exhibit the canonical transcriptomic Hsp90 β client signature, because their dependence on Hsp90 β is regulated post-translationally rather than transcriptionally.

3.5 Disease Validation

Disease validation was performed using 96 proteins measured in human cardiac proteomics comparing non-failing (NF) to failing ischemic cardiomyopathy (ICM) hearts (n=6 per group). T-test analysis identified 3 proteins significantly dysregulated at $p < 0.05$:

Q9UKX3 (logFC = +2.686, $p = 0.034$), P01859 (logFC = +0.811, $p = 0.044$), and P62277 (logFC = +0.619, $p = 0.011$). Of the 46 confirmed Hsp90 β clients, 7 were detected in the cardiac dataset (UniProt IDs: Q02218, P38646, P11177, P05091, P25705, P06576, P11498). However, none of these 7 overlapping proteins were among the 3 significantly dysregulated ICM proteins. Fisher's exact test yielded $p = 1.0$ (odds ratio = 0.0), indicating no significant enrichment of Hsp90 β clients among ICM-dysregulated proteins.

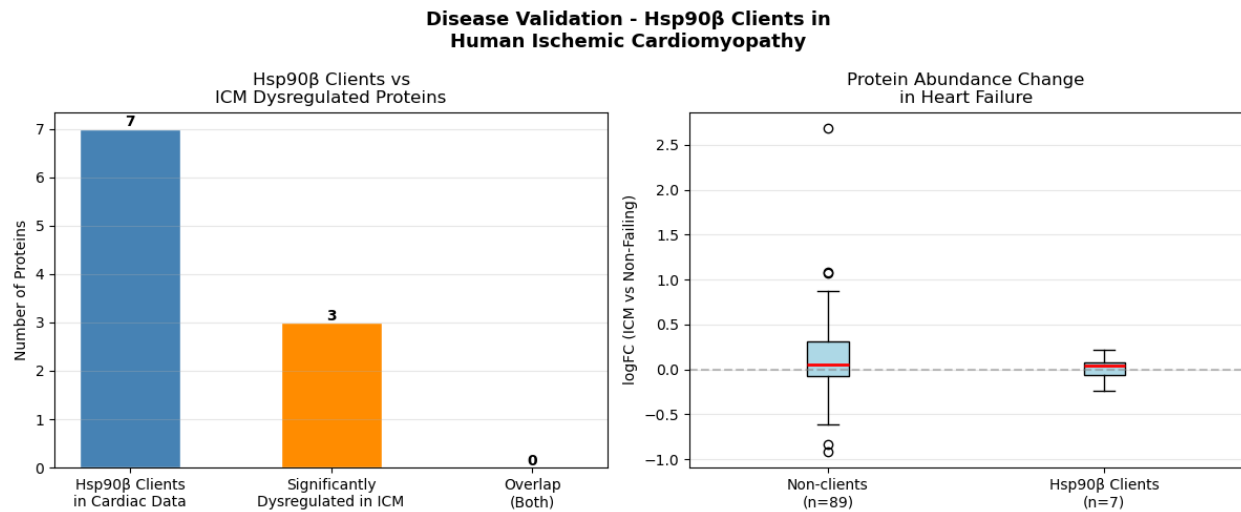


Fig2: Disease Validation – Hsp90AB1 clients in Human Ischemic Cardiomyopathy

4. Discussion

4.1 Model Performance and Hypothesis Testing

Hypothesis 1 was partially supported. The SVM achieved ROC-AUC = 0.740 ± 0.095 in nested cross-validation, significantly above random chance as confirmed by permutation test ($p = 0.0099$). However, the 0.80 AUC threshold set in the hypothesis was not reached. This moderate performance likely reflects two compounding limitations: the small positive training set (46 proteins, approximately 9 per test fold in 5-fold CV) and the exclusive use of transcriptomic features, which miss post-translationally regulated clients such as HAX-1 and CypD.

The SVM outperformed the Random Forest by approximately 8.5% absolute AUC (0.740 vs. 0.654), consistent with the well-established advantage of kernel-based methods over ensemble tree methods in small, high-dimensional, imbalanced classification problems.

The SVM identifies an optimal hyperplane defined by a small number of support vectors, requiring fewer positive examples than the Random Forest, which needs sufficient data in each leaf node of each tree.

The Neural Network performed below random chance ($AUC = 0.456 \pm 0.111$), confirming that deep learning is inappropriate for this dataset size. An MLP with a hidden layer of 64 neurons has approximately 384 learnable parameters, while only 37 positive examples were available per training fold. The high fold-to-fold variance (± 0.111) indicates the network learned inconsistent and non-generalizable patterns from the small positive set. This is a valid and expected negative result rather than a model failure.

4.2 Biological Significance of Feature Importance

The finding that $\alpha VS \beta_{logFC}$ (32%) is the strongest predictor of Hsp90 β client status is the most novel and biologically significant result of this project. This feature captures isoform specificity whether a protein responds differently when Hsp90 β is removed versus when Hsp90 α is removed. When Hsp90 β is knocked out, the cell upregulates Hsp90 α as a compensatory mechanism, as directly demonstrated in Lam Laboratory western blots. Proteins that specifically depend on Hsp90 β cannot be rescued by this compensatory Hsp90 α upregulation, resulting in distinct expression patterns between the two knockout conditions. The model's reliance on this feature computationally validates that the defining characteristic of a true Hsp90 β client is not simply responsiveness to Hsp90 in general, but beta-isoform specificity that resists alpha rescue.

The secondary importance of βKO_{logFC} (27%) and βKO_{logCPM} (25%) further supports the biological coherence of the model. Proteins downregulated when Hsp90 β is genetically absent and highly expressed in cardiac tissue are more likely to be functionally dependent on Hsp90 β for their stability. The minimal contribution of FDR values (6% and 10% combined) is expected statistical confidence is a measurement quality metric, not a biological signal.

4.3 HAX-1 and CypD: Post-Translational Regulation

The low Hsp90 β client probability scores for HAX-1 (1.3%) and CypD (1.4%) are not model failures but rather biologically informative results. Both proteins showed near-zero RNA-level changes in the Beta KO (FDR = 1.0 for both), indicating their dependence on Hsp90 β is entirely post-translational. Hsp90 β physically stabilizes HAX-1 and CypD as a chaperone without affecting the transcription of their genes. The model, trained exclusively on transcriptomic features, correctly recognizes that these proteins lack the canonical transcriptomic client signature. This finding motivates future work incorporating protein-level measurements (proteomics, co-IP quantification) as additional features that would allow the model to capture post-translational Hsp90 β dependence.

4.4 Disease Validation Limitations

Hypothesis 2 was not supported. However, the negative result is most likely attributable to insufficient statistical power rather than a genuine absence of disease relevance. The cardiac validation dataset contained only 96 proteins measured in 6 patients per group, yielding only 3 significantly dysregulated proteins at $p < 0.05$. With such a small denominator, detecting significant enrichment of any protein subset is statistically near-impossible (Fisher's exact test has minimal power with these cell counts). The existing wet lab evidence from the Lam Laboratory demonstrating that Hsp90 β ablation reduces infarct size and preserves mitochondrial structure in I/R injury provides strong biological validation of disease relevance independent of this computational analysis.

4.5 Methodological Contributions

A key methodological contribution of this work is the rigorous handling of data leakage. Two categories of leakage were identified and addressed. First, pharmacological response features (Increase with Val, Decrease with Gel) were excluded as model features because they were used to define the positive labels, which would create circular reasoning. Second, molecular weight was excluded after observing that it produced a suspicious AUC of 1.0 driven by size bias rather than genuine learning. The use of nested

cross-validation, recommended by the course instructor following Varma and Simon (2006), further ensured that hyperparameter tuning did not contaminate performance estimates. The permutation test confirmed that the final reported performance reflects genuine biological signal ($p = 0.0099$).

5. References

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